

Layout-Aware Markdown Reconstruction

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Abstract

We evaluate whether multimodal models can recover document reading order, table structure, and inline visual references from born-digital PDFs.

The proposed metric, Doc2MD-F, combines block alignment with visual fact checks.

Keywords: document parsing; Markdown; multimodal evaluation

1. Introduction

Faithful document conversion is different from question answering. A model can answer a local question while omitting most of the document.

Our benchmark asks for a single Markdown artifact, not a summary.

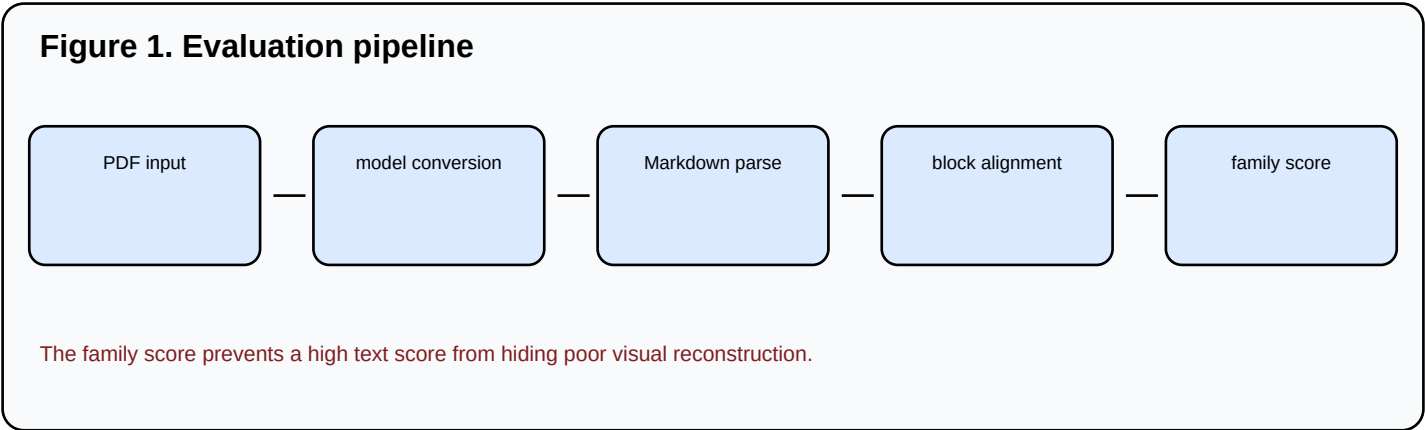
Footnote 1: We treat repeated page numbers as optional boilerplate.

2. Method

The block alignment score is defined as:

$$$$$ = 0.25T + 0.20R + 0.20Q + 0.10F + 0.10V + 0.10M + 0.05C$$$$$

where T is text fidelity, R is reading order, Q is table quality, F is form quality, V is visual fact quality, M is math/code quality, and C is Markdown cleanliness.



3. Results

Model	Text	Tables	Visuals	Overall
Model A	93.1	71.4	64.0	81.2
Model B	90.5	88.2	52.5	82.6
Model C	86.0	79.0	76.5	80.5

Model B has the highest overall score but the lowest visual score.

References

[1] Clark et al. Structured document evaluation. 2024.
[2] Lin et al. Table grid similarity. 2025.
[3] Nair and Gomez. Visual fact extraction for charts. 2026.